

The Bergman shift has exactly two reducing subspaces

A disproof of conjecture 00000000930

Independent mathematical submission

Abstract

For the ordinary unweighted Bergman shift, the closed complex reducing subspaces are exactly $\{0\}$ and the whole Bergman space. Their cardinality is therefore 2, not the cardinality of the continuum. We prove this directly on square-integrable holomorphic functions by division by $z - w$. The accompanying Lean 4 project also proves the analytic completeness bridge, the equivalence of two standard reducing-subspace definitions, and the exact cardinality statement.

1 Statement, conventions, and conclusion

The supplied English statement is:

Definition: The Bergman shift (M_z on A^2). Conjecture: The lattice of reducing subspaces of the Bergman shift has the cardinality of the continuum, each indexed by Blaschke products (a complete classification of reducing subspaces).

The Chinese statement specifies the same operator and the same continuum-cardinality and Blaschke-indexing assertions.

Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ and let $dA = dx dy$ be unnormalized planar Lebesgue area. Set

$$A^2(\mathbb{D}) = \left\{ f : \mathbb{D} \rightarrow \mathbb{C} : f \text{ is holomorphic and } \int_{\mathbb{D}} |f(z)|^2 dA(z) < \infty \right\}.$$

Our inner product is conjugate-linear in the first variable:

$$\langle f, g \rangle = \int_{\mathbb{D}} \overline{f(z)} g(z) dA(z).$$

Using dA/π instead multiplies this inner product by a positive constant and leaves its topology, orthogonality, and reducing subspaces unchanged. The operator in question is exactly

$$S = M_z, \quad (Sf)(z) = zf(z).$$

It is bounded, with $\|Sf\|_2 \leq \|f\|_2$.

A reducing subspace means a closed complex linear subspace $M \subseteq A^2(\mathbb{D})$ for which $S(M) \subseteq M$ and $S(M^\perp) \subseteq M^\perp$. Equivalently, it is a closed complex linear subspace invariant under both S and its Hilbert-space adjoint S^* ; this equivalence is proved below and in Lean.

Theorem 1. *The complete collection of closed complex reducing subspaces of M_z on $A^2(\mathbb{D})$ is $\{\{0\}, A^2(\mathbb{D})\}$. In particular, its cardinality is exactly 2, and is not the cardinality $\mathfrak{c}$ of the continuum.*

This disproves a necessary, explicit clause of the conjecture and hence disproves the conjecture as a conjunction. The Blaschke-product sentence does not specify an indexing map or an equivalence relation on labels. No such map or equivalence relation is invented here; the disproof is the failure of the continuum-cardinality clause for the actual operator.

2 The analytic Hilbert-space bridge

For completeness we give the analytic facts required to apply orthogonal projection to every closed subspace in the statement. These facts are also proved in the accompanying project, rather than supplied as hypotheses.

Lemma 2 (Area mean value). *If f is holomorphic on a neighborhood of $\overline{B(w, r)}$ and $r > 0$, then*

$$\int_{B(w, r)} f(z) dA(z) = \pi r^2 f(w).$$

Proof. Cauchy's integral formula, with $z = w + \rho e^{i\theta}$, gives

$$\int_0^{2\pi} f(w + \rho e^{i\theta}) d\theta = 2\pi f(w) \quad (0 < \rho \leq r).$$

Integrating radially and using polar coordinates and Fubini gives

$$\int_{B(w, r)} f dA = \int_0^r \rho \int_{-\pi}^{\pi} f(w + \rho e^{i\theta}) d\theta d\rho = \pi r^2 f(w).$$

Changing the angular interval does not change the integral because the integrand is 2π -periodic. Continuity on the closed disk makes the polar integrand integrable on its compact parameter rectangle. Translation handles an arbitrary center w . $\square$

Lemma 3 (Evaluation estimate). *Write $C = \|1\|_{L^2(\mathbb{D}, dA)}$, a fixed finite constant. If $\overline{B(w, r)} \subset \mathbb{D}$ and $r > 0$, then every $f \in A^2(\mathbb{D})$ satisfies*

$$|f(w)| \leq \frac{C}{\pi r^2} \|f\|_2.$$

Proof. The area mean-value formula, the triangle inequality, and Cauchy–Schwarz give

$$\pi r^2 |f(w)| \leq \int_{B(w, r)} |f| dA \leq \int_{\mathbb{D}} |f| dA \leq C \|f\|_2.$$

The final inequality is the real L^2 inner-product inequality applied to 1 and $|f|$. $\square$

Proposition 4. *The natural inclusion of $A^2(\mathbb{D})$ into $L^2(\mathbb{D}, dA)$ is injective and has closed range. Consequently $A^2(\mathbb{D})$ is a Hilbert space with the displayed inner product.*

Proof. Two continuous functions on the open disk that agree almost everywhere agree everywhere: if their difference were nonzero at a point, continuity would make it nonzero on a nonempty open set of positive area. This proves injectivity.

Suppose $f_n \in A^2(\mathbb{D})$ converge to u in L^2 . They are L^2 -Cauchy. Fix $0 < t < 1$ and put $r = (1 - t)/2 > 0$. For every $w \in B(0, t)$, $\overline{B(w, r)} \subset \mathbb{D}$. Lemma 3 applied to $f_m - f_n$ therefore gives

$$\sup_{w \in B(0, t)} |f_m(w) - f_n(w)| \leq \frac{C}{\pi r^2} \|f_m - f_n\|_2.$$

Thus (f_n) is uniformly Cauchy on every smaller disk. It has a pointwise limit g on $\mathbb{D}$, with locally uniform convergence. The locally uniform limit theorem for holomorphic functions makes g holomorphic.

Convergence in L^2 implies convergence in measure, so some subsequence converges almost everywhere to a representative of u . The same subsequence converges pointwise on $\mathbb{D}$ to g . Uniqueness of limits gives $g = u$ almost everywhere. Hence g is square integrable and represents u , proving that the range is closed. Completeness follows from completeness of L^2 . $\square$

3 Local division and the commutant

Lemma 5 (Division inside A^2). *For $f \in A^2(\mathbb{D})$ and $w \in \mathbb{D}$, define*

$$q(z) = \begin{cases} \frac{f(z) - f(w)}{z - w}, & z \neq w, \\ f'(w), & z = w. \end{cases}$$

Then $q \in A^2(\mathbb{D})$ and

$$f - f(w)1 = (S - wI)q.$$

Proof. The removable singularity theorem makes q holomorphic. Its continuity at w gives numbers $r > 0$ and $C_0 \geq 0$ such that $|q(z)| \leq C_0$ for $|z - w| < r$. Outside that neighborhood,

$$|q(z)| \leq r^{-1}(|f(z)| + |f(w)|).$$

Thus everywhere on the disk,

$$|q(z)| \leq C_0 + r^{-1}(|f(z)| + |f(w)|).$$

The right side is in L^2 , since $f \in L^2$ and the disk has finite area. Therefore $q \in A^2(\mathbb{D})$. The stated identity holds pointwise, including at w . $\square$

Lemma 6. *Let $P : A^2(\mathbb{D}) \rightarrow A^2(\mathbb{D})$ be any complex-linear map satisfying $PS = SP$. Then*

$$(Pf)(w) = f(w)(P1)(w) \quad (f \in A^2(\mathbb{D}), w \in \mathbb{D}).$$

Proof. Apply P to the identity in Lemma 5 and commute it with $S - wI$:

$$Pf - f(w)P1 = (S - wI)Pq.$$

Evaluation at w annihilates the right side. $\square$

No boundedness assumption on P is used in this lemma.

Proposition 7. *If $P : A^2(\mathbb{D}) \rightarrow A^2(\mathbb{D})$ is complex linear, $PS = SP$, and $P^2 = P$, then $P = 0$ or $P = I$.*

Proof. Put $h = P1$. By Lemma 6, P acts pointwise as multiplication by h . The equation $P^2 1 = P1$ gives

$$h(w)(h(w) - 1) = 0 \quad (w \in \mathbb{D}).$$

Both h and $h - 1$ are holomorphic, and the disk is connected. The identity principle therefore implies that h vanishes identically or that $h - 1$ vanishes identically. The pointwise formula for Pf then gives $P = 0$ or $P = I$, respectively. $\square$

4 Every closed reducing subspace and the exact cardinality

First verify the equivalence of reducing-subspace conventions. If M is closed, Proposition 4 gives $M^{\perp\perp} = M$. If $S(M^\perp) \subseteq M^\perp$, then for $x \in M$ and $y \in M^\perp$,

$$\langle y, S^*x \rangle = \langle Sy, x \rangle = 0,$$

so $S^*x \in M^{\perp\perp} = M$. Conversely, if $S^*(M) \subseteq M$, then for $x \in M^\perp$ and $y \in M$,

$$\langle y, Sx \rangle = \langle S^*y, x \rangle = 0,$$

so $Sx \in M^\perp$. Together with $S(M) \subseteq M$, these are exactly the two stated definitions.

Proof of Theorem 1. Let M be any closed reducing subspace. The Hilbert projection theorem provides the orthogonal direct sum

$$A^2(\mathbb{D}) = M \oplus M^\perp.$$

Let P be projection onto the first summand. For $u \in M$ and $v \in M^\perp$, the two invariances give

$$PS(u + v) = Su = SP(u + v).$$

Thus P is a complex-linear idempotent commuting with S . Proposition 7 implies $P = 0$ or $P = I$, hence $M = \{0\}$ or $M = A^2(\mathbb{D})$.

Both subspaces are closed and reducing. They are distinct because the constant function 1 belongs to $A^2(\mathbb{D})$ and is not zero. Sending false to $\{0\}$ and true to $A^2(\mathbb{D})$ is consequently a bijection from the two-element Boolean type onto the entire collection of reducing subspaces. The cardinality is exactly 2. Since every finite cardinal is strictly smaller than $\mathfrak{c}$, it is not $\mathfrak{c}$. $\square$

5 Formalization and reproducibility

The portable project is in `lean/`. It pins Lean 4.19.0 and Mathlib v4.19.0, at commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`

Run `lake build` from that directory. The project-level option `warningAsError = true` makes warnings fatal. All included project modules are reachable from the default target.

The definitions preserve the original analytic objects:

- `Bergman930.space` consists of actual complex functions, holomorphic on `disk` and in `MemLp` at exponent 2 for restricted planar Lebesgue measure. They are zero outside the disk solely to choose a unique representative of a function defined on $\mathbb{D}$.
- `toL2_injective` and `functionEquivL2` identify those functions exactly with `l2Space`, the range in genuine area- L^2 . `inner_toL2` proves the displayed integral inner-product formula. `l2Space_isClosed` proves closedness and supplies its completeness.
- `shift` is pointwise multiplication by z ; `bergmanShift` is this same operator transported to `l2Space`, with its contraction bound checked. The reducing predicate uses this space's inherited L^2 topology and orthogonal complement.

The principal checked statements, all in namespace `Bergman930`, are:

Mathematical assertion	Lean declaration
Local division	<code>division_identity</code>
Commuting maps act pointwise	<code>commuting_map_apply</code>
Commuting idempotents are trivial	<code>commuting_idempotent_eq_zero_or_id</code>
Area mean value	<code>disk_mean</code>
L^2 evaluation estimate	<code>evaluation_bound</code>
Actual Bergman completeness	<code>l2Space_isClosed</code>
Adjoint-definition equivalence	<code>isReducing_iff_adjoint</code>
All closed reducing subspaces	<code>reducing_eq_bot_or_top</code>
Exact cardinality 2	<code>reducing_cardinality</code>
Failure of continuum cardinality	<code>not_continuum</code>

In particular, the final theorem is

`Cardinal.mk Bergman930.ReducingSubspace ≠ Cardinal.continuum.`

The preceding exact-cardinality theorem has right side 2. The type `ReducingSubspace` is a subtype of *all* complex submodules of the genuine Bergman Hilbert space satisfying `IsReducing`; it is not an enumeration used as a definition.

`Verification.lean` prints the concrete definitions, the principal theorem types, and their axiom dependencies. Fresh compilation of all owned source modules succeeds with warnings treated as errors. The inspected statements depend only on the standard Lean axioms `propext`, `Classical.choice`, and `Quot.sound`. There are no custom axioms, unfinished proofs, `native_decide` calls, or unsafe or partial proof dependencies in the submitted source.

The trust boundary is explicit: the supplied precompiled objects of the pinned standard dependencies are trusted imports; every owned source module is freshly elaborated and checked. Three standard Mathlib modules absent from that local precompiled cache were also freshly compiled from unchanged pinned source. The final portable project uses their canonical Mathlib imports and includes no renamed copies or dependency cache. Exact commands, exit codes, raw outputs, and source hashes accompany the verification record. No numerical computation is needed for the disproof.